Open with an external PDF viewer for optimal hyperlinks.

PayFlow Control

Data Analytics Case Study

Product	: PayFlow Control
Product Type	: B2B SaaS MVP Prototype
Positioning	: Payment Operations Intelligence SaaS
Project Type	: Independent Product Development Portfolio Project
Analysis Type	: Payment Transaction Data Analysis
Data Source	: Public Payment Transaction Dataset
Tools	: PostgreSQL, Power BI

Table of Contents

Phase 1 — Dataset Profiling

1. Executive Purpose

The project begins by establishing whether the available transaction data is sufficiently suitable for payment operations, risk analysis, dashboard development, and product discovery. The analysis is deliberately framed from a product and operations perspective rather than as a pure fraud-detection exercise.

2. Dataset Overview

Dataset	Records	Role
transactions_train.csv	300,113	Primary analytical dataset
transactions_test.csv	99,887	Independent validation dataset
Combined	400,000	Analytical population

The analyzed transaction data covers 1 January–30 December 2023. One record represents one payment transaction.

3. Analytical Coverage

- Transaction activity and value
- Customer and merchant behaviour
- Payment channel and device behaviour
- Velocity and geographic indicators
- Risk and fraud analysis
- Product-oriented operational insight

4. Product Capability Coverage

Capability	Data Support	Boundary
Payment Visibility	High	Transaction attributes directly support monitoring
Risk Monitoring	High	Multiple risk and behavioural indicators available
Exception Management	Medium	Requires transparent business rules
Compliance Support	Medium	KYC and operational indicators provide support
Reconciliation Management	Limited	Settlement fields are unavailable in the public dataset

The dataset is therefore suitable for demonstrating transaction, risk, and operational analytics, while reconciliation and some compliance capabilities require explicit simulation or product assumptions.

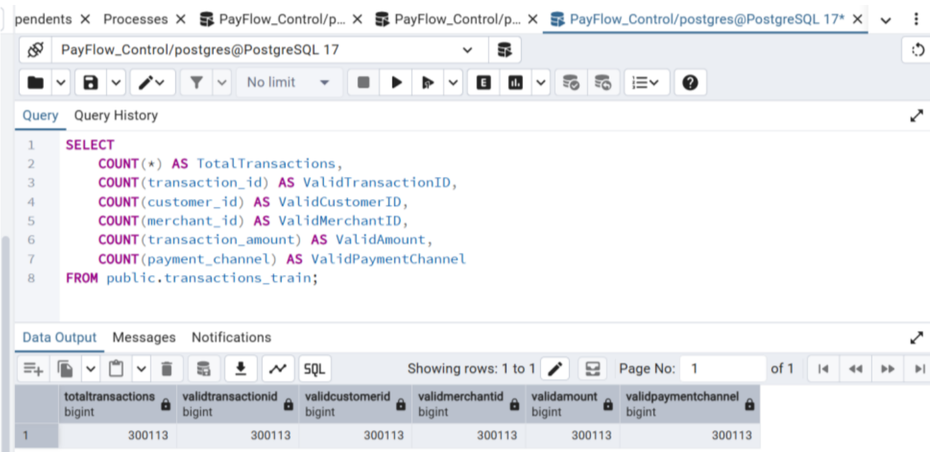
Phase 2 — Data Quality Assessment

1. Quality Framework

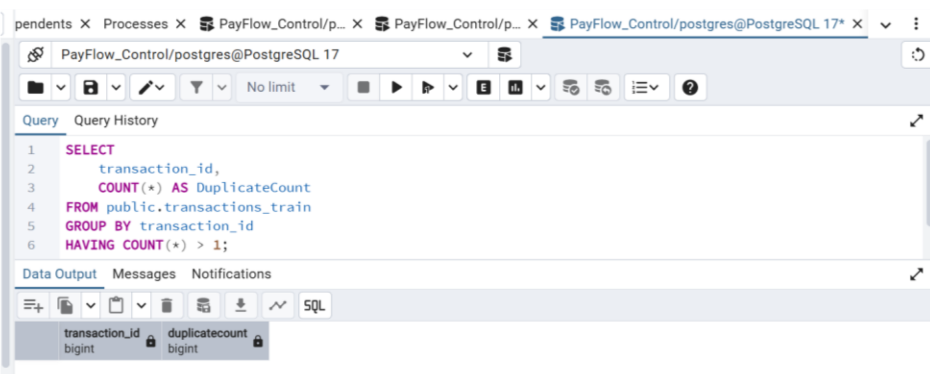
Before analysis, the dataset is assessed for completeness, uniqueness, validity, consistency, proxy accuracy, and integrity. Because this is public data, business accuracy cannot be externally verified; logical and analytical validation are used instead.

2. Core SQL Validation

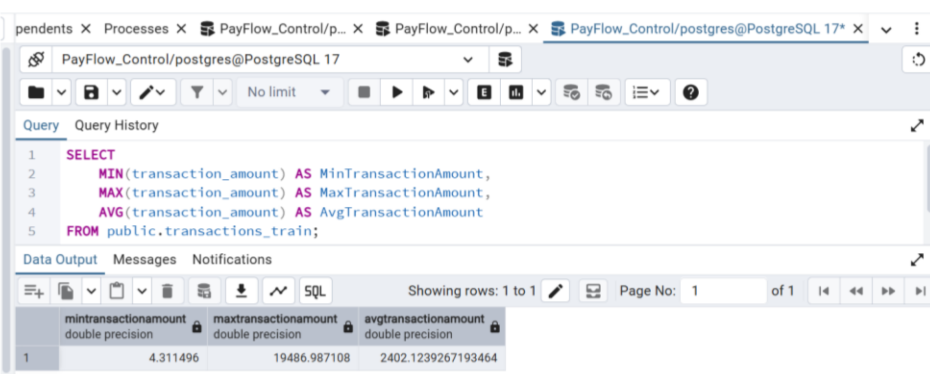
Completeness / Null Validation



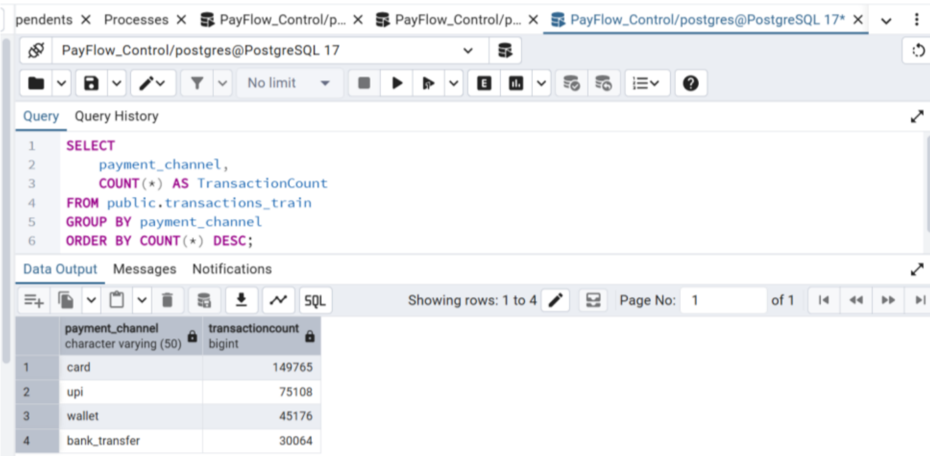
Duplicate Transaction_id



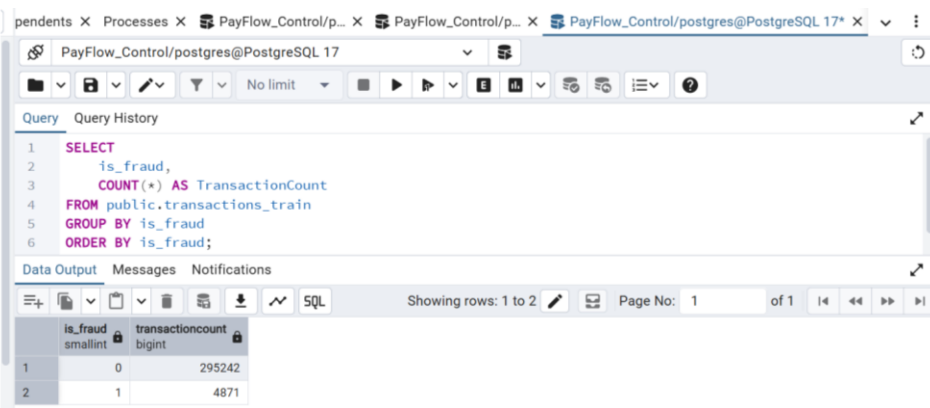
Transaction Amount Range



Payment Channel Distribution



Fraud Distribution



3. Quality Interpretation

Validation focuses on whether data quality could distort transaction volume, payment value, fraud rates, risk metrics, or dashboard segmentation. Outliers are not automatically removed because unusually large or unusual payments can be legitimate operational events.

4. Preparation Principle

Every transformation is documented and must have a business purpose. Raw data remains traceable, while derived analytical fields are created in the preparation layer.

Phase 3 — Data Preparation & Feature Engineering

1. Analytical Architecture

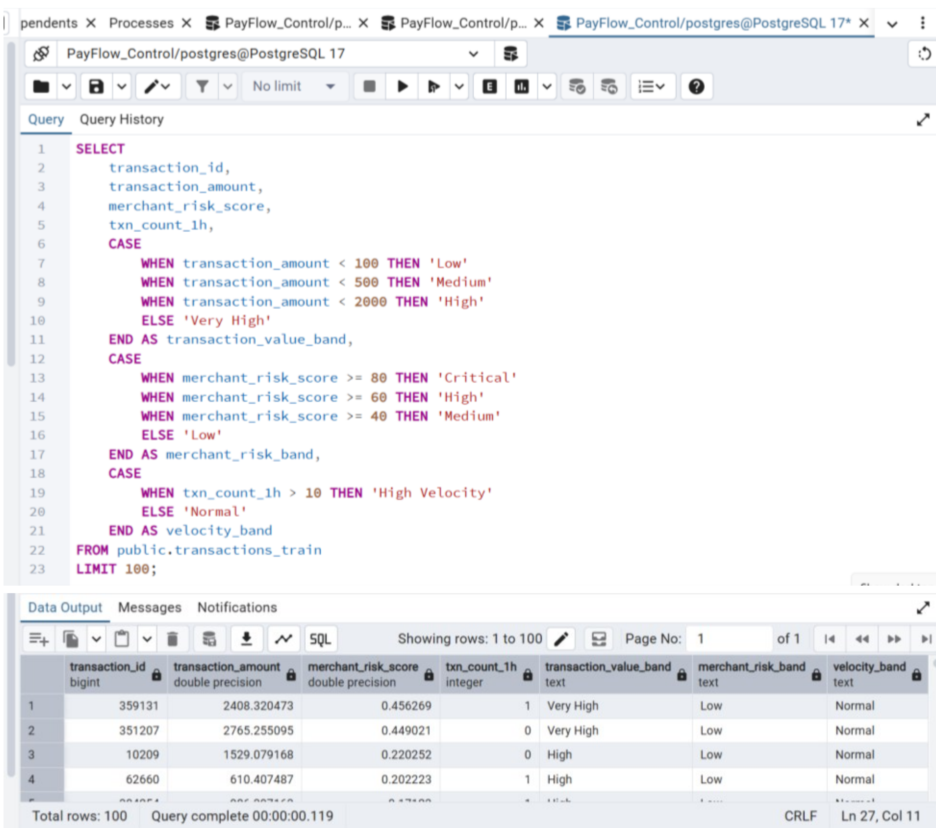
The preparation model follows a simple layered approach:

Raw Dataset → Clean Analytical Dataset → Business Feature Layer → Power BI Semantic Model → Dashboard

2. Business Features

Domain	Representative Derived Features
Transaction	Value Band, Hour, Day, Weekend, Business Hours
Customer	Maturity, Spending Segment, Risk Profile, KYC Category
Merchant	Risk Band, Priority, Monitoring Group
Behaviour	High Velocity Flag, Frequent Failure Flag, Behaviour Alert
Risk	Risk Category, Risk Priority, Investigation Level
Operations	High Value, International, Large Deviation, Investigation Candidate

3. Key Transformation Logic



```sql
SELECT
    transaction_id,
    transaction_amount,
    merchant_risk_score,
    txn_count_1h,
    CASE
        WHEN transaction_amount < 100 THEN 'Low'
        WHEN transaction_amount < 500 THEN 'Medium'
        WHEN transaction_amount < 2000 THEN 'High'
        ELSE 'Very High'
    END AS transaction_value_band,
    CASE
        WHEN merchant_risk_score >= 80 THEN 'Critical'
        WHEN merchant_risk_score >= 60 THEN 'High'
        WHEN merchant_risk_score >= 40 THEN 'Medium'
        ELSE 'Low'
    END AS merchant_risk_band,
    CASE
        WHEN txn_count_1h > 10 THEN 'High Velocity'
        ELSE 'Normal'
    END AS velocity_band
FROM public.transactions_train
LIMIT 100;
```

	transaction_id bigint	transaction_amount double precision	merchant_risk_score double precision	txn_count_1h integer	transaction_value_band text	merchant_risk_band text	velocity_band text
1	359131	2408.320473	0.456269	1	Very High	Low	Normal
2	351207	2765.255095	0.449021	0	Very High	Low	Normal
3	10209	1529.079168	0.220252	0	High	Low	Normal
4	62660	610.407487	0.202223	1	High	Low	Normal

Total rows: 100 Query complete 00:00:00.119

4. Power BI Preparation

Time dimensions include year, quarter, month, week, day, hour, weekday, weekend, and business-hours indicators. These dimensions support consistent KPI and dashboard calculations.

Phase 4 — Exploratory Data Analysis

1. Analytical Objective

EDA establishes the descriptive baseline: transaction scale, value distribution, temporal activity, payment channels, customer and merchant behaviour, risk indicators, and fraud occurrence.

2. Representative SQL Validation

Executive Transaction Overview

```sql
SELECT
    COUNT(*) AS TotalTransactions,
    SUM(transaction_amount) AS TotalValue,
    AVG(transaction_amount) AS AverageValue
FROM public.transactions_train;
```

	totaltransactions bigint	totalvalue double precision	averagevalue double precision
1	300113	720908618.0195228	2402.123926719345

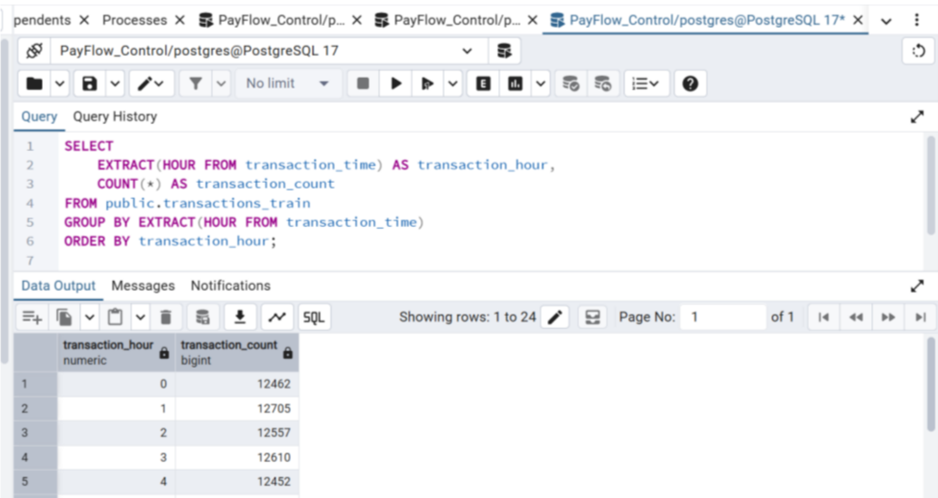
Transaction Value Distribution

```sql
SELECT CASE
        WHEN transaction_amount < 100 THEN 'Low'
        WHEN transaction_amount < 500 THEN 'Medium'
        WHEN transaction_amount < 2000 THEN 'High'
        ELSE 'Very High'
    END AS transaction_value_band, COUNT(*) AS transaction_count
FROM public.transactions_train GROUP BY
    CASE
        WHEN transaction_amount < 100 THEN 'Low'
        WHEN transaction_amount < 500 THEN 'Medium'
        WHEN transaction_amount < 2000 THEN 'High'
        ELSE 'Very High'
    END;
```

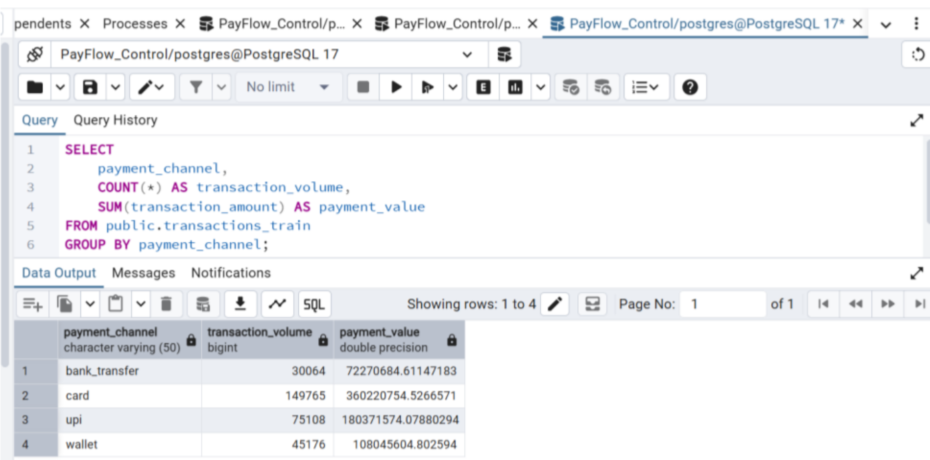
	transaction_value_band text	transaction_count bigint
1	High	128987
2	Very High	151395
3	Low	1021
4	Medium	18710

Total rows: 4 Query complete 00:00:00.425

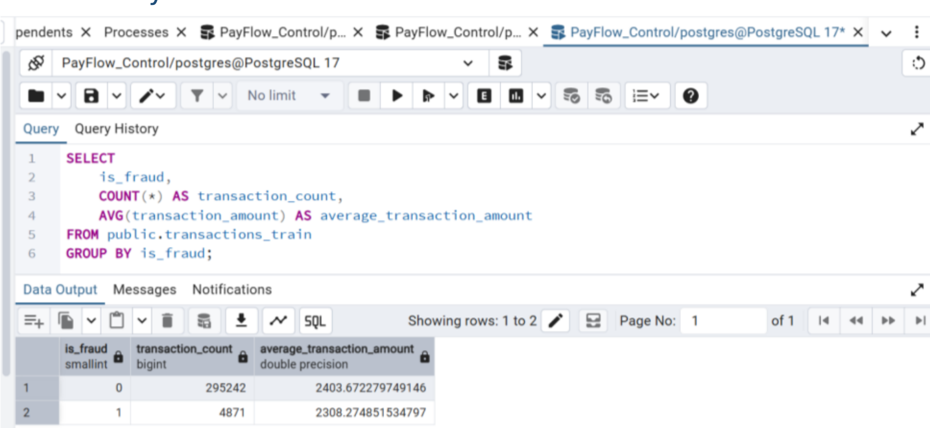
Time-Based Analysis



Payment Channel Analysis



Fraud Analysis



3. Analytical Outputs

- Executive transaction overview
- Transaction volume and value analysis

- Time-based transaction analysis
- Payment channel analysis
- Customer and merchant behaviour
- Initial risk analysis
- Initial fraud analysis
- Exploratory relationship assessment

The EDA stage provides the descriptive evidence base for the more focused risk and operational findings used later in this case study.

Phase 5 — Advanced Business & Behavioural Analytics

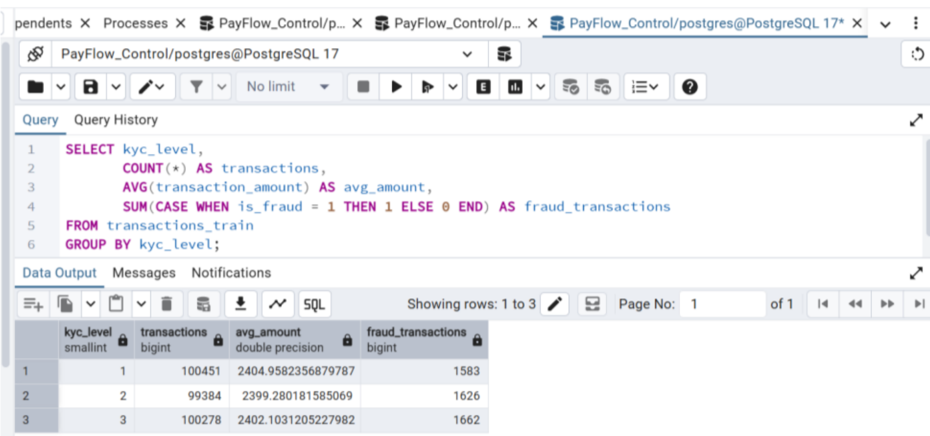
1. Analytical Framework

Each analytical workstream follows a consistent evidence-to-decision flow: **Business Question → Analytical Method → SQL Validation → Power BI Visualization → Business Insight → Product Recommendation.**

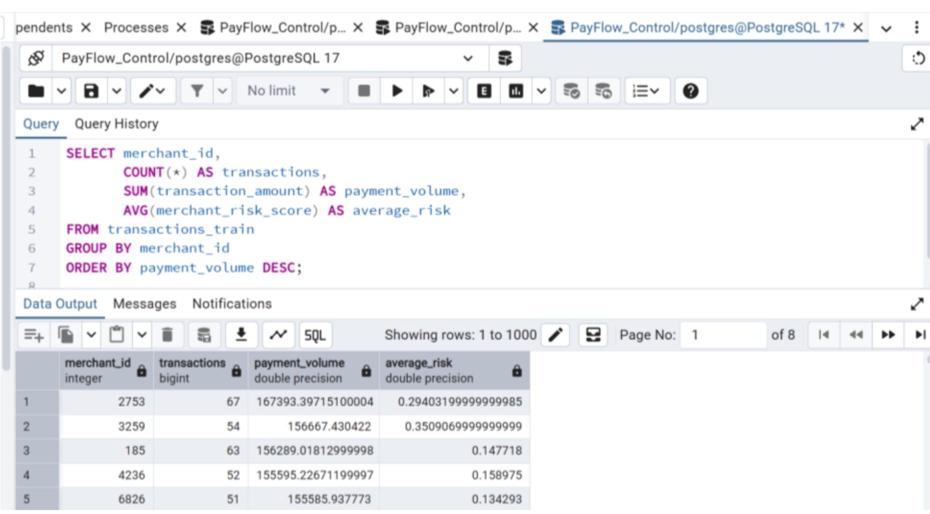
The SQL queries in this phase provide the analytical validation layer for five complementary workstreams: **Customer Behaviour, Merchant Behaviour, Transaction Velocity, Payment Channel, and Risk Indicators**. These analyses extend beyond descriptive EDA by examining behavioural patterns, operational risk signals, and relationships that can inform PayFlow Control's product capabilities.

2. Analytical Workstreams & SQL Validation

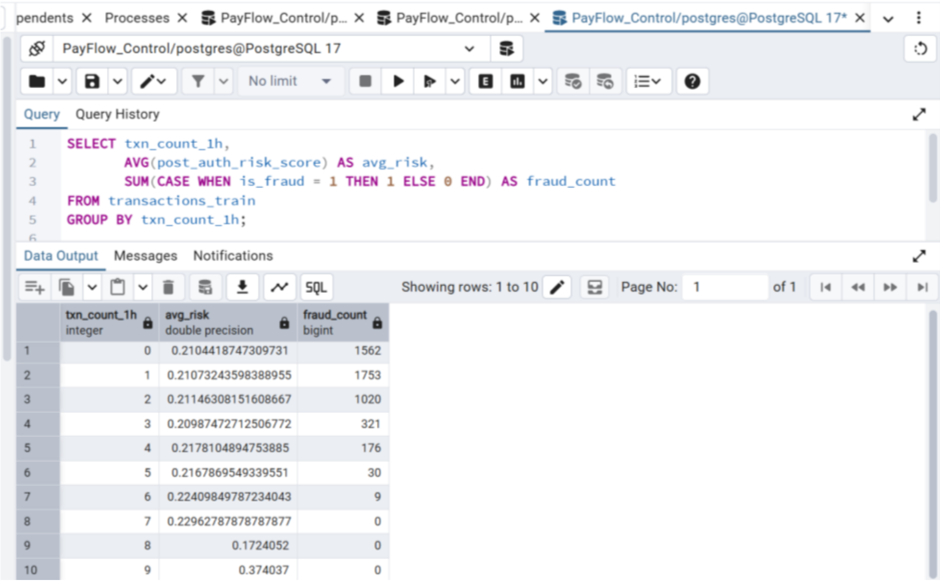
Customer Behaviour



Merchant Behaviour



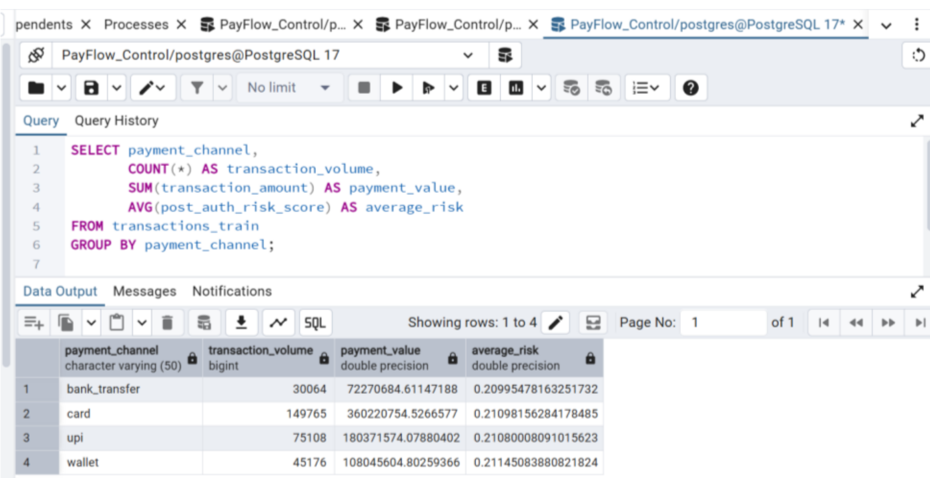
Velocity



```sql
SELECT txn_count_1h,
       AVG(post_auth_risk_score) AS avg_risk,
       SUM(CASE WHEN is_fraud = 1 THEN 1 ELSE 0 END) AS fraud_count
FROM transactions_train
GROUP BY txn_count_1h;
```

	txn_count_1h integer	avg_risk double precision	fraud_count bigint
1	0	0.2104418747309731	1562
2	1	0.21073243598388955	1753
3	2	0.21146308151608667	1020
4	3	0.20987472712506772	321
5	4	0.2178104894753885	176
6	5	0.2167869549339551	30
7	6	0.22409849787234043	9
8	7	0.22962787878787877	0
9	8	0.1724052	0
10	9	0.374037	0

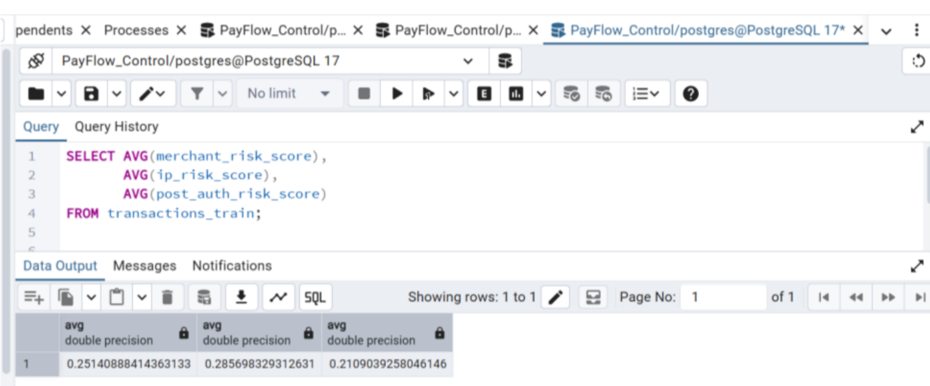
Payment Channel



```sql
SELECT payment_channel,
       COUNT(*) AS transaction_volume,
       SUM(transaction_amount) AS payment_value,
       AVG(post_auth_risk_score) AS average_risk
FROM transactions_train
GROUP BY payment_channel;
```

	payment_channel character varying (50)	transaction_volume bigint	payment_value double precision	average_risk double precision
1	bank_transfer	30064	72270684.61147188	0.20995478163251732
2	card	149765	360220754.5266577	0.21098156284178485
3	upi	75108	180371574.07880402	0.21080008091015623
4	wallet	45176	108045604.80259366	0.21145083880821824

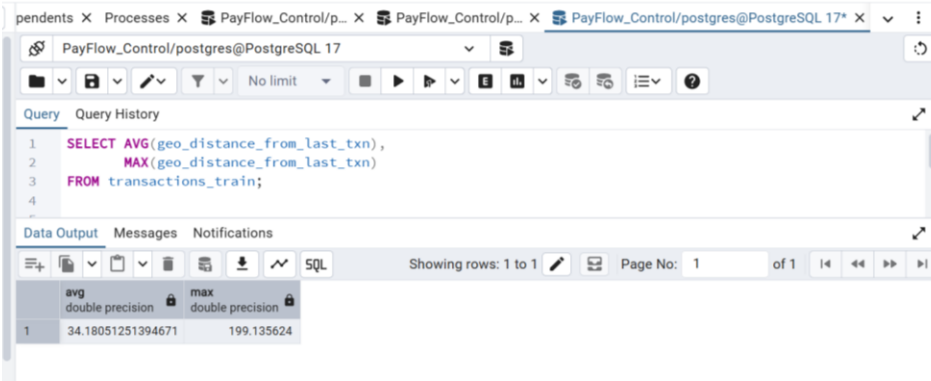
Risk Indicators



```sql
SELECT AVG(merchant_risk_score),
       AVG(ip_risk_score),
       AVG(post_auth_risk_score)
FROM transactions_train;
```

	avg double precision	avg double precision	avg double precision
1	0.25140888414363133	0.285698329312631	0.2109039258046146

Geographic Behaviour



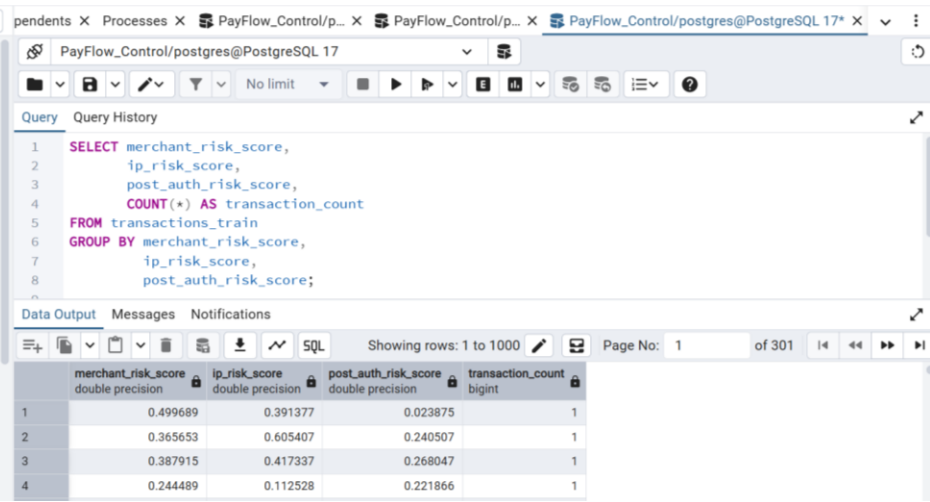
These analyses establish candidate operational signals, but only findings subsequently confirmed in the final Power BI dashboard are treated as actual evidence in this portfolio.

Phase 6 — Risk, Fraud & Operational Intelligence

1. Purpose

This phase converts descriptive and behavioural signals into explainable operational intelligence. It does not attempt to build a machine-learning fraud model. The objective is to support investigation prioritization and operational monitoring.

Multi-Factor Risk Assessment



The analytical principle is that risk should be interpreted using multiple contextual indicators rather than a single score.

2. Operational Exception Logic

The dataset does not contain explicit exception records. Therefore, operational exceptions are treated as simulated business rules rather than observed fields.

Simulated Exception	**Rule**
High Velocity	Transaction count exceeds threshold
High Amount Deviation	Transaction materially exceeds customer average
Elevated Merchant Risk	Merchant risk exceeds operational threshold
Cross-Border Review	International transaction with elevated risk
Multiple Failed Attempts	Failed transaction count exceeds threshold

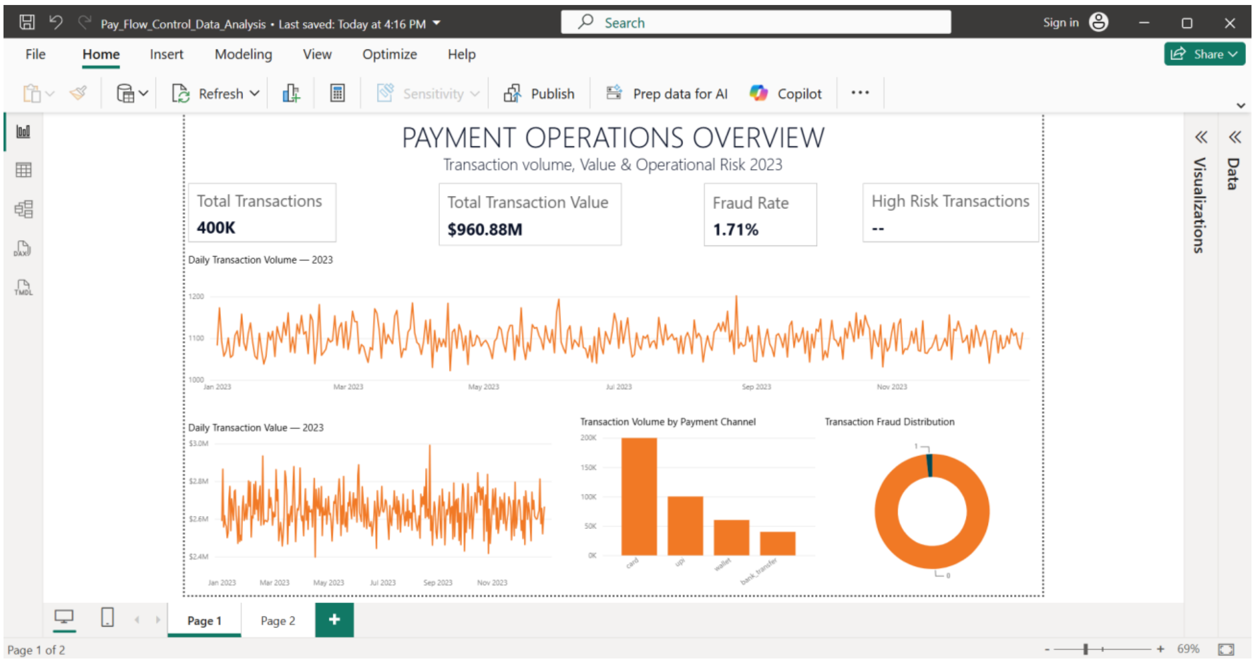
3. Compliance Support

KYC coverage, high-risk transaction ratio, international transaction ratio, investigation coverage, and operational review status can support compliance-readiness visibility. They are not presented as evidence of regulatory compliance.

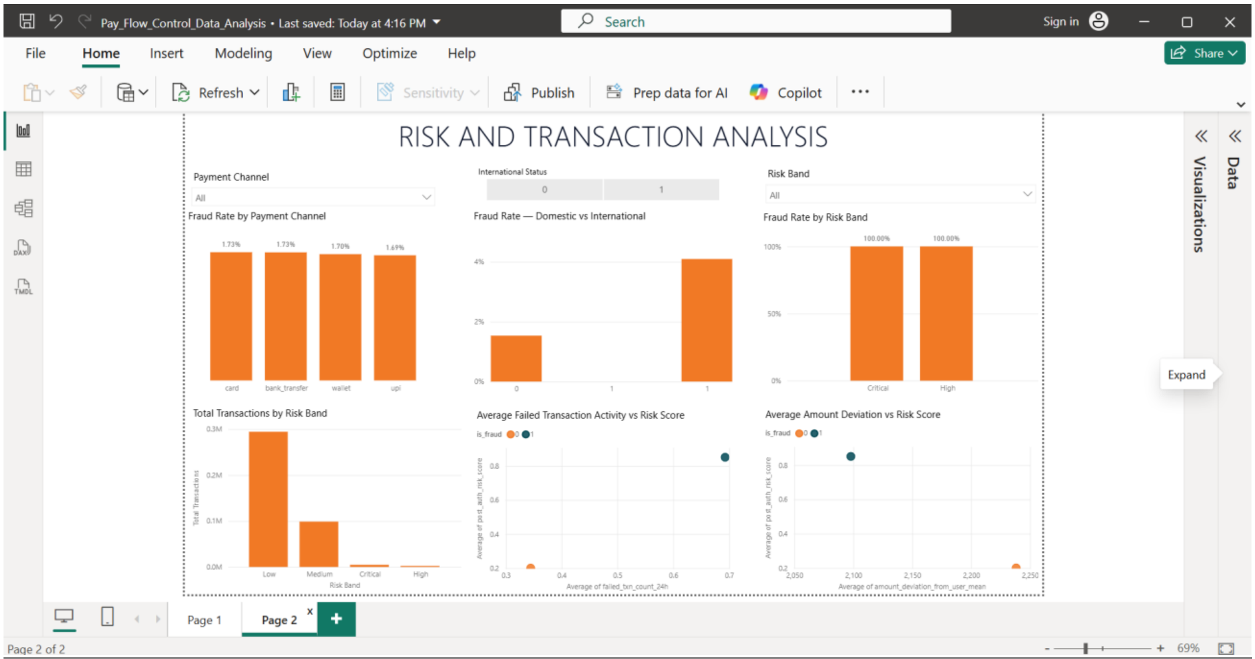
4. Final Power BI Analysis — Evidence Used for Product Findings Dashboard Scope

The final Power BI implementation was intentionally reduced to two pages so that the dashboard remains recruiter-friendly and decision-oriented.

Page	Purpose
Page 1 — Payment Operations Overview	Executive KPIs, transaction activity, payment performance, and overall operational visibility
Page 2 — Risk & Transaction Analysis	Fraud exposure, risk bands, failed-transaction activity, channel comparison, and amount-deviation analysis



This screenshot is the primary visual evidence for the overall transaction and payment-operations context.



This screenshot is the primary visual evidence for the data-driven findings in Phase 8 of the product case study.

5. Evidence Boundary

The final dashboard findings are based on the analyzed public transaction dataset. They are external evidence for product discovery, not customer evidence.

6. Data-Driven Findings — Final Power BI Results

Finding 1 — International Transactions Show Higher Fraud Incidence

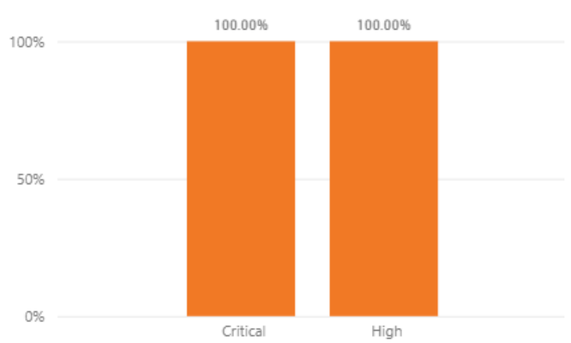
Observation	:	International transactions show an observed fraud rate of approximately 4.10%, compared with approximately 1.54% for domestic transactions.
Insight	:	International transaction status is a meaningful risk-segmentation variable in this dataset, with fraud incidence approximately 2.7× higher for international transactions.
Product implication	:	PayFlow Control should support international transaction monitoring, risk segmentation, and investigation prioritization.

Fraud Rate — Domestic vs International

Finding 2 — Risk Bands Strongly Separate Fraud Outcomes

Observation	:	After calibrating risk-band thresholds to the 0–1 scale used by the analyzed risk score, Low and Medium bands show 0% observed fraud, while High and Critical show 100% observed fraud.
Insight	:	The risk score provides very strong separation between fraudulent and non-fraudulent transactions in this dataset.
Product implication	:	Risk indicators, risk-band filtering, and high-risk transaction visibility are appropriate supporting capabilities.
Evidence boundary	:	The 100% result is dataset-specific and must not be generalized to real-world risk bands.

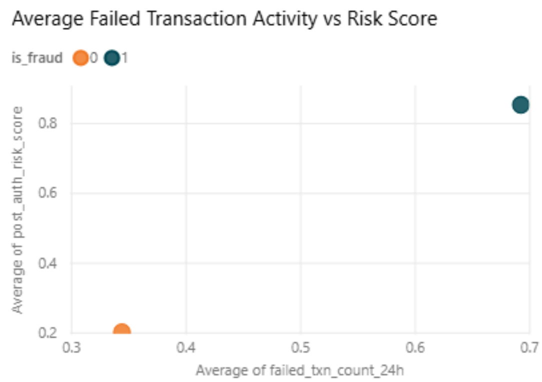
Fraud Rate by Risk Band



7. Data-Driven Findings — Operational Signals

Finding 3 — Fraudulent Transactions Show Higher Failed-Transaction Activity

Observation	:	Average failed-transaction activity is approximately 0.69 for fraudulent transactions versus approximately 0.34 for non-fraudulent transactions.
Insight	:	Fraudulent transactions show roughly twice the failed-transaction activity in this dataset.
Product implication	:	Failed-transaction activity can be incorporated as an operational risk indicator and investigation signal.

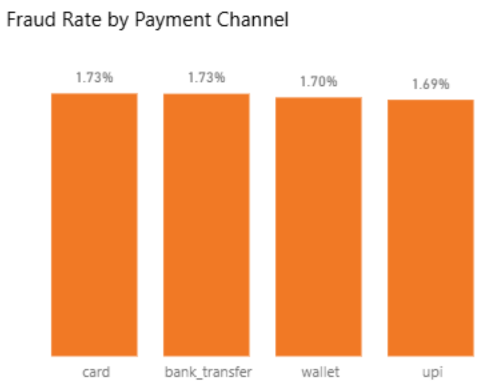


Finding 4 — Payment Channel Alone Shows Limited Differentiation

Observation : Fraud rates are relatively similar across payment channels, approximately 1.69%–1.73%.

Insight : Payment channel alone does not materially differentiate fraud exposure in this dataset.

Product implication : Channel should remain a monitoring dimension, but not the primary standalone risk trigger.

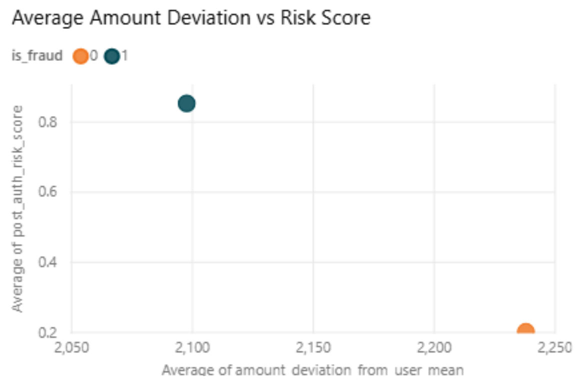


Finding 5 — Amount Deviation Is Not a Strong Standalone Fraud Signal

Observation : Average amount deviation is approximately 2,238 for non-fraudulent transactions and 2,098 for fraudulent transactions.

Insight : Amount deviation from the user's historical mean does not show a strong standalone fraud signal in this dataset.

Product implication : It should not be prioritized as an independent risk trigger; it may be more useful when combined with other signals.



8. Data-to-Product Decision Framework

From Evidence to Product

The central analytical principle is: Data → Insight → Product Decision. The purpose of the analysis is not to maximize the number of charts, but to identify evidence that changes or validates product priorities.

Data Observation	Insight	Product Decision
International fraud rate ~2.7× domestic	International status is useful risk segmentation	Add international monitoring/filtering
High/Critical risk bands show 100% observed fraud	Risk score strongly separates outcomes in this dataset	Prioritize risk indicators and high-risk visibility
Fraud transactions show ~2× failed activity	Repeated failures may support investigation	Include failed activity as a risk signal
Channel fraud rates are similar	Channel alone is weak differentiation	Use channel as a monitoring dimension
Amount deviation shows limited differentiation	Weak standalone signal	Do not prioritize as an independent trigger

9. What the Analysis Does Not Prove

- It does not prove that all hospitality businesses have the same fraud patterns.
- It does not validate customer willingness to pay.
- It does not prove settlement or reconciliation pain because settlement data is unavailable.
- It does not establish regulatory compliance.
- It does not constitute a production fraud model.

This evidence discipline is important because the project is an independent portfolio case study rather than a commercial customer engagement.

10. KPI Framework & Measurement

Executive KPIs

KPI	Purpose
Total Transactions	Operational throughput
Total Payment Value	Payment volume
Average Transaction Value	Transaction economics
Fraud Rate	Observed risk exposure
International Transaction Ratio	Cross-border activity
High-Risk Transaction Ratio	Monitoring workload

Operational & Risk KPIs

- Transaction Success Ratio
- Failed Transaction Rate
- High-Value Transaction Rate
- Average Risk Score
- High-Risk Transaction Count
- Investigation Queue Size (future/simulated)

KPI Design Standard

Each KPI should have a business definition, formula, SQL validation, DAX measure where applicable, dashboard placement, business owner, and decision supported. This creates traceability from source data through Power BI to product decisions.

Representative SQL KPI Validation

11. Power BI Design & Analytical Communication

Dashboard Principles

- Each page answers a specific business question.
- KPIs appear before detailed diagnostics.
- Charts are limited to decision-relevant dimensions.
- Filters support investigation without creating visual clutter.
- Actual findings are separated from assumptions and simulated scenarios.

Final Dashboard Structure

Page	Key Content
Page 1	Executive KPIs, transaction activity, payment performance, operational overview
Page 2	Fraud by domestic/international status, fraud by risk band, failed activity, channel fraud, amount deviation and risk relationships

SQL-to-Power BI Traceability

SQL is used as a reproducible validation layer, while Power BI provides the decision-facing visualization. The final dashboard is intentionally smaller than the original analytical blueprint because the portfolio prioritizes clarity and evidence over chart volume.

12. Product Recommendations & Portfolio Interpretation

Recommendations Supported by Evidence

Evidence	Product Capability
Higher international fraud incidence	Cross-border transaction monitoring and filtering
Strong separation by risk band	Risk indicators and high-risk transaction visibility
Higher failed activity among fraud transactions	Operational failure/risk signal
Limited channel differentiation	Channel monitoring rather than channel-specific risk rules
Weak standalone amount-deviation signal	Use as contextual rather than primary trigger

13. Capabilities Retained as Hypotheses

- Reconciliation Management
- Exception Management beyond simulated rules
- Compliance Support workflows
- Customer validation and workflow urgency
- Commercial willingness to pay

These areas remain strategically relevant to PayFlow Control, but the current public dataset cannot independently validate them. They should therefore be supported through assumptions, simulation, future research, or additional operational data.

14. Positioning Implication

The findings support PayFlow Control as a Payment Operations Intelligence SaaS rather than a dedicated fraud-management platform. Risk monitoring is one operational control capability within a broader payment-operations layer.

15. Executive Conclusion

This case study demonstrates how a public transactional dataset can be transformed into product-relevant analytical evidence through structured profiling, data-quality assessment, feature engineering, exploratory analysis, behavioural analysis, risk analysis, SQL validation, and Power BI visualization.

The final Power BI analysis identifies several actionable signals. International transactions show materially higher observed fraud incidence than domestic transactions. Risk bands provide strong separation in the analyzed dataset. Fraudulent transactions also show higher failed-transaction activity. Conversely, payment channel and amount deviation provide limited standalone differentiation.

These results do not constitute customer validation or a production fraud model. Their value is in demonstrating disciplined analytical reasoning: identifying what the data supports, separating evidence from assumptions, and translating observations into product decisions.

16. Capability Demonstrated

Area	Evidence in Case Study
Data Analytics	Profiling, quality assessment, EDA, behavioural and risk analysis
SQL	Validation, aggregation, segmentation and reproducible metric logic
Power BI	Executive dashboard and risk/transaction analysis
Product Management	Data-to-insight-to-product decision translation
Business Analysis	Operational interpretation and prioritization
Analytical Governance	Clear distinction between evidence, simulation and hypothesis

The overall outcome is a concise, evidence-informed analytics case study that supports the broader PayFlow Control product portfolio without presenting simulated or public-data findings as real customer experience.

17. Overall Product Relevance

The combined findings from the public transaction dataset provide evidence that the operational problems addressed by PayFlow Control are relevant for further product development. The analysis identified meaningful transaction activity, payment-channel variation, operational exceptions, transaction-risk patterns, fraud indicators, and areas requiring greater operational visibility. These findings collectively support the relevance of developing PayFlow Control as a **Payment Operations Intelligence SaaS**, with capabilities spanning **Payment Visibility, Reconciliation Management, Exception Management, Risk Monitoring, and Compliance Support**.

The findings do not validate commercial demand, willingness to pay, or customer adoption. Instead, they provide **data-driven external evidence that the proposed product problem space and core capability areas are sufficiently relevant to justify continued product development and customer validation**.